Set 6 Problem 1: Let $a, b \in \Delta - \{0\}$ and consider the twice punctured disks $D_1 = \Delta - \{0, a\}$ and $D_2 = \Delta - \{0, b\}$. Find a necessary and sufficient condition for D_1 and D_2 to be biholomorphic and determine all biholomorphic maps $f : D_1 \to D_2$.

Firstly, we find our necessary condition. Suppose $f : D_1 \to D_2$ is a biholomorphic function. Then as $|f(z)| \leq 1$ for all $z \in D_1$, we know that f has a removable singularity at 0 and a. Hence, we may define a holomorphic extension $\widetilde{f}$ of f to the entire unit disk. Also note that $\widetilde{f}(\Delta) \subseteq \overline{\Delta}$ and is an open set by the open map theorem. It follows that $\widetilde{f}$ is a holomorphic function from Δ to Δ.

Suppose for the sake of contradiction that there exists $c \in \Delta - \{0\}$ such that $\widetilde{f}(0) = \widetilde{f}(c) = \alpha$.

Then pick disjoint disks Δ_0 and Δ_c centered at 0 and c. By the open mapping theorem, we can find a disk Δ_α centered at α such that $\Delta_\alpha \subseteq \widetilde{f}(\Delta_0)$ and $\Delta_\alpha \subseteq \widetilde{f}(\Delta_c)$. Pick $\beta \in \Delta_\alpha - \{\alpha, \widetilde{f}(a)\}$. Then there exists $u \in \Delta_0 - \{0\}$ and $v \in \Delta_c - \{c\}$ such that $\widetilde{f}(u) = \beta = \widetilde{f}(v)$. Importantly, neither u nor v equals 0 or a. So, u and v are in D_1. And this contradicts that f is injective on D_1.

By analogous reasoning, we can conclude there doesn't exist $c \in \Delta - \{a\}$ such that $\widetilde{f}(a) = \widetilde{f}(c)$. Consequently, we can conclude that:

$$\widetilde{f} \in \mathrm{Aut}(\Delta) \text{ and } \{\widetilde{f}(0), \widetilde{f}(a)\} = \{0, b\}$$

Now any automorphism of Δ has the form $e^{i\theta}\frac{z - z_0}{1 - \overline{z}_0 z}$ where z_0 the element of Δ that is mapped to 0 and $\theta \in \mathbb{R}$. Hence, we have two cases:

- If $\widetilde{f}(0) = 0$ then we know $\widetilde{f}(z) = e^{i\theta}z$ for some $\theta \in \mathbb{R}$. Also $b = \widetilde{f}(a) = e^{i\theta}a$ implies that $\frac{b}{a} = e^{i\theta}$. This shows that $|a| = |b|$ and that $\widetilde{f}(z) = \frac{b}{a}z$.

- If $\widetilde{f}(a) = 0$ then we know $\widetilde{f}(z) = e^{i\theta}\frac{z - a}{1 - \overline{a}z}$ for some $\theta \in \mathbb{R}$. Also, $b = \widetilde{f}(0) = e^{i\theta}\frac{-a}{1}$ implies that $\frac{-b}{a} = e^{i\theta}$. This shows that $|a| = |b|$ and that $\widetilde{f}(z) = \frac{-b}{a} \cdot \frac{z - a}{1 - \overline{a}z}$.

This proves that if D_1 and D_2 are biholomorphic then we have that $|a| = |b|$ and the only two biholomorphisms are $z \mapsto \frac{b}{a}z$ and $z \mapsto \frac{-b}{a} \cdot \frac{z - a}{1 - \overline{a}z}$.

To finish off, this is also a sufficient condition. If $|a| = |b|$ then both of the above mappings are biholomorphisms of the unit disk satisfying that $\{0, a\}$ is mapped to $\{0, b\}$. ■

We can also characterize the biholomorphisms of the upper half plane (which I'll denote as $\mathfrak{h}^+$).

Note that any biholomorphism of $\mathfrak{h}^+$ can be written as $c^{-1} \circ f \circ c$ where $c : \mathfrak{h}^+ \to \Delta$ is the Cayley transform and f is a biholomorphism of Δ.

Consider any matrix $\left[\begin{smallmatrix} A & B \\ \overline{B} & \overline{A} \end{smallmatrix}\right] \in \mathrm{SU}(1,1)$ associated with f and let $\left[\begin{smallmatrix} 1 & -i \\ 1 & i \end{smallmatrix}\right]$ and $\frac{1}{2}\left[\begin{smallmatrix} 1 & 1 \\ i & -i \end{smallmatrix}\right]$ be the matrices associated with c and c^{-1} respectively. Then the matrix associated with $c^{-1} \circ f \circ c$ is $\left[\begin{smallmatrix} \mathrm{Re}(A)+\mathrm{Re}(B) & \mathrm{Im}(A)-\mathrm{Im}(B) \\ -\mathrm{Im}(A)-\mathrm{Im}(B) & \mathrm{Re}(A)-\mathrm{Re}(B) \end{smallmatrix}\right] \in \mathrm{GL}_2(\mathbb{R})$. Also, we scaled the matrices associated with c and c^{-1} so that they'd have determinants $2i$ and $\frac{-i}{2}$ respectively. Therefore, our real matrix has determinant 1.

Combining this with an algebra homework exercise (see *page 286*), we can conclude that the set of biholomorphisms of $\mathfrak{h}^+$ is isomorphic to $\mathrm{PSL}_2(\mathbb{R})$.

One other note to quickly make is that there is no biholomorphism from $\mathbb{C}$ to Δ (We know this by Liouville's theorem).

Set 6 Problem 2: The pseudo-hyperbolic distance on Δ is given by $d(z,w) = \left|\frac{z-w}{1-\overline{z}w}\right|$.

(a) Show by direct calculation that if f is a biholomorphism of Δ then
$$d(z,w) = d(f(z), f(w)).$$

Recall that any biholomorphism of f is a composition of a rotation $z \mapsto e^{i\theta}z$ (where $\theta \in \mathbb{R}$) and another function of the form $\varphi_a(z) = \frac{z-a}{1-\overline{a}z}$ where $a \in \Delta$. Therefore, it suffices to show that rotations and functions of the form φ_a preserve pseudo-hyperbolic distance.

- $d(e^{i\theta}z, e^{i\theta}w) = \left|\frac{e^{i\theta}z - e^{i\theta}w}{1-e^{-i\theta}\overline{z}e^{i\theta}w}\right| = |e^{i\theta}| \cdot \left|\frac{z-w}{1-1\overline{z}w}\right| = 1 \cdot d(z,w)$

- $$\begin{aligned} d(\varphi_a(z), \varphi_a(w)) &= \left|\frac{\frac{z-a}{1-\overline{a}z} - \frac{w-a}{1-\overline{a}w}}{1-\left(\frac{\overline{z}-\overline{a}}{1-a\overline{z}}\right)\left(\frac{w-a}{1-\overline{a}w}\right)}\right| \\ &= \left|\frac{\frac{(1-\overline{a}w)(z-a)-(1-\overline{a}z)(w-a)}{(1-\overline{a}z)(1-\overline{a}w)}}{\frac{(1-a\overline{z})(1-\overline{a}w)-(\overline{z}-\overline{a})(w-a)}{(1-a\overline{z})(1-\overline{a}w)}}\right| \\ &= \left|\frac{1-a\overline{z}}{1-\overline{a}z}\right| \cdot \left|\frac{1-\overline{a}w}{1-\overline{a}w}\right| \cdot \left|\frac{(1-\overline{a}w)(z-a)-(1-\overline{a}z)(w-a)}{(1-a\overline{z})(1-\overline{a}w)-(\overline{z}-\overline{a})(w-a)}\right| \\ &= 1 \cdot 1 \cdot \left|\frac{z-a-\overline{a}wz+|a|^2w-w+a+\overline{a}wz-|a|^2z}{1-\overline{a}w-a\overline{z}+|a|^2w\overline{z}-w\overline{z}+a\overline{z}+\overline{a}w-|a|^2}\right| \\ &= \left|\frac{z+|a|^2w-w-|a|^2z}{1+|a|^2w\overline{z}-w\overline{z}-|a|^2}\right| = \left|\frac{(z-w)(1-|a|^2)}{(1-\overline{z}w)(1-|a|^2)}\right| = \left|\frac{z-w}{1-\overline{z}w}\right| = d(z,w) \end{aligned}$$

(b) Let $f : \Delta \to \Delta$ be holomorphic. Using (a) to reduce to a familiar case, show that $d(f(z), f(w)) \leq d(z,w)$. Thus holomorphic maps contract pseudo-hyperbolic distance.

Consider the biholomorphisms φ_{-w} and $\varphi_{f(w)}$ of the unit disk. Then upon setting $\alpha = \varphi_w(z)$, we have by part (a) that:
$$\begin{aligned} d(f(z), f(w)) = d((\varphi_{f(w)} \circ f)(z), (\varphi_{f(w)} \circ f)(w)) &= d((\varphi_{f(w)} \circ f)(z), 0) \\ &= d((\varphi_{f(w)} \circ f \circ \varphi_{-w})(\alpha), 0) \\ &= \left|\frac{(\varphi_{f(w)} \circ f \circ \varphi_{-w})(\alpha) - 0}{1-0}\right| \end{aligned}$$

Yet now as $\varphi_{f(w)} \circ f \circ \varphi_{-w}$ is a holomorphism from Δ to Δ sending 0 to 0, we know by Schwarz' lemma that $|(\varphi_{f(w)} \circ f \circ \varphi_{-w})(\alpha)| \leq |\alpha|$. Hence, we may conclude that $d(f(z), f(w)) \leq |\alpha| = |\varphi_w(z)| = \frac{|z-w|}{|1-\overline{w}z|} = \frac{|z-w|}{|1-\overline{z}w|} = d(z,w)$.

(c) Suppose $f : \Delta \to \Delta$ is a holomorphic function and that there exists distinct $z, w \in \Delta$ with $d(f(z), f(w)) = d(z, w)$. Show that f is a biholomorphism and consequently by (a) that $d(f(z), f(w)) = d(z, w)$ for all $z, w \in \Delta$.

Like in part (b), write:

$$d(f(z), f(w)) = |(\varphi_{f(w)} \circ f \circ \varphi_{-w})(\varphi_w(z))| \leq |\varphi_w(z)| = d(z, w).$$

This time however, note that because $|(\varphi_{f(w)} \circ f \circ \varphi_{-w})(\varphi_w(z))| = |\varphi_w(z)|$ (since $d(f(z), f(w)) = d(z, w)$) and $\varphi_w(z) \neq 0$ (since $z \neq w$), we must have by Schwarz' lemma that $\varphi_{f(w)} \circ f \circ \varphi_{-w}$ is a rotation R. It follows that $f = \varphi_{-f(w)} \circ R \circ \varphi_w$ is a biholomorphism.

(d) Show that d is a metric.

Firstly, it's obvious that $d(z, w) = 0$ iff $z = w$ and that $d(z, w) = d(w, z)$ (just take the complex conjugate of the denominator...). Thus, the only hard part is to show that d satisfies the triangle inequality. In other words, we want to show that if $z, w, s \in \Delta$ then $d(z, s) \leq d(z, w) + d(w, s)$.

To start off, by applying φ_s to z, w, s and invoking part (a), we can assume without loss of generality that $s = 0$. Hence, it suffices to prove that $d(z, 0) \leq d(z, w) + d(w, 0)$ for all $z, w \in \Delta$.

Next, by applying a rotation in Δ to $z, w, 0$ and then invoking part (a), we can assume without loss of generality that z is a nonnegative real number. Hence, we may now write $z = r_1$ and $w = r_2 e^{i\theta}$ where $r_1, r_2 \in [0, 1)$ and $\theta \in \mathbb{R}$. In turn:

- $d(z, 0) = r_1$.
- $d(z, w) + d(w, 0) = \left|\frac{r_1 - r_2 e^{i\theta}}{1 - r_1 r_2 e^{i\theta}}\right| + r_2$.

Note that if $r_2 = 0$ then $|d(z, w)| = |\frac{r_1 - 0}{1 - 0}| = r_1$ and so our inequality trivially holds. So, we may assume $r_2 \neq 0$. Also note that if $r_2 \geq r_1$, then we are already done. So, we may assume now that $r_2 < r_1$ (which happens to give us that $r_2, r_1 \neq 0$). In turn we know that $r_1 - r_2 > 0$. So, our desired inequality is equivalent to saying that:

$$(r_1 - r_2)^2 \leq \left|\frac{r_1 - r_2 e^{i\theta}}{1 - r_1 r_2 e^{i\theta}}\right|^2 = \frac{\left|r_1 - r_2 e^{i\theta}\right|^2}{\left|1 - r_1 r_2 e^{i\theta}\right|^2} = \frac{(r_1 - r_2 e^{i\theta})(r_1 - r_2 e^{-i\theta})}{(1 - r_1 r_2 e^{i\theta})(1 - r_1 r_2 e^{-i\theta})}$$

$$= \frac{r_1^2 - 2r_1 r_2 \mathrm{Re}(e^{i\theta}) + r_2^2}{1 - 2r_1 r_2 \mathrm{Re}(e^{i\theta}) + r_1^2 r_2^2}$$

Therefore, it suffices to show that $(r_1 - r_2)^2 \leq \frac{r_1^2 - 2r_1 r_2 \cos(\theta) + r_2^2}{1 - 2r_1 r_2 \cos(\theta) + r_1^2 r_2^2}$.

Or in other words:

$$(r_1^2 - 2r_1 r_2 + r_2^2)(1 - 2r_1 r_2 \cos(\theta) + r_1^2 r_2^2) \leq r_1^2 - 2r_1 r_2 \cos(\theta) + r_2^2$$

Upon expanding and canceling, we get that:

$$(-2r_1 r_2 + r_1^4 r_2^2 - 2r_1^3 r_2^3 + r_1^2 r_2^4) - 2r_1 r_2 (r_1 - r_2)^2 \cos(\theta) \leq -2r_1 r_2 \cos(\theta)$$

Then by dividing by $r_1 r_2$ and rearranging a little more:

$$(-2 + r_1^3 r_2 - 2r_1^2 r_2^2 + r_1 r_2^3) \leq 2((r_1 - r_2)^2 - 1) \cdot \cos(\theta)$$

Finally, the right side is minimized when $\cos(\theta) = +1$. So, we need to show that $(-2 + r_1^3 r_2 - 2r_1^2 r_2^2 + r_1 r_2^3) \leq 2((r_1 - r_2)^2 - 1)$. Fortunately, this is equivalent to saying that:

$$\begin{aligned} 0 &\leq 2(r_1 - r_2)^2 - (r_1^3 r_2 - 2r_1^2 r_2^2 + r_1 r_2^3) \\ &= 2(r_1 - r_2)^2 - r_1 r_2 (r_1^2 - 2r_1 r_2 + r_2^2) = 2(r_1 - r_2)^2 - r_1 r_2 (r_1 - r_2)^2 \\ &= (2 - r_1 r_2)(r_1 - r_2)^2 \end{aligned}$$

Then as $r_1, r_2 \in (0, 1)$ and $r_2 < r_1$, we know that $(2 - r_1 r_2) > 0$ and $(r_1 - r_2) > 0$. So this last inequality holds and we are done.

(e) As an application to (b), assume $f : \Delta \to \Delta$ is a holomorphic function satisfying that $f(\frac{1}{2}) = \frac{1}{4}$. Show that $\frac{1}{21} \leq |f(\frac{1}{3})| \leq \frac{9}{19}$.

Firstly, note that:

$$\left| \frac{\frac{1}{4} - f(\frac{1}{3})}{1 - \frac{1}{4} f(\frac{1}{3})} \right| = d(\tfrac{1}{4}, f(\tfrac{1}{3})) \leq d(\tfrac{1}{2}, \tfrac{1}{3}) = \left| \frac{\frac{1}{2} - \frac{1}{3}}{1 - \frac{1}{6}} \right| = \frac{1/6}{5/6} = \frac{1}{5}$$

It follows that $5|\frac{1}{4} - f(\frac{1}{3})| \leq |1 - \frac{1}{4} f(\frac{1}{3})|$. By triangle inquality and reverse triangle inquality, we get that:

$$5\left|\tfrac{1}{4} - |f(\tfrac{1}{3})|\right| \leq 1 + \tfrac{1}{4}|f(\tfrac{1}{3})|$$

Case 1: Suppose $|f(\frac{1}{3})| \leq \frac{1}{4}$.

Then $\frac{5}{4} - 5|f(\frac{1}{3})| \leq 1 + \frac{1}{4}|f(\frac{1}{3})| \implies \frac{1}{4} \leq \frac{21}{4}|f(\frac{1}{3})| \implies \frac{1}{21} \leq |f(\frac{1}{3})|$

Case 2: Suppose $|f(\frac{1}{3})| \geq \frac{1}{4}$.

Then $5|f(\frac{1}{3})| - \frac{5}{4} \leq 1 + \frac{1}{4}|f(\frac{1}{3})| \implies \frac{19}{4}|f(\frac{1}{3})| \leq \frac{9}{4} \implies |f(\frac{1}{3})| \leq \frac{9}{19}$.

(f) Let $f : \Delta \to \Delta$ be holomorphic and satisfy that $f(\frac{1}{3}) = \frac{1}{2}$. Show that f has no zeros in the disc $\{z : |z| < \frac{1}{7}\}$.

Note that if $f(w) = 0$ for some $w \in \Delta$, then by part (b) we must have that:

$$\tfrac{1}{2} = d(\tfrac{1}{2}, 0) \leq d(\tfrac{1}{3}, w) = \left| \frac{\frac{1}{3} - w}{1 - \frac{w}{3}} \right| = \left| \frac{1 - 3w}{3 - w} \right| \implies |3 - w| \leq 2|1 - 3w|$$

Therefore, to show that f has no zeros in the disk $\{z : |z| < \frac{1}{7}\}$, it suffices to show that there is some $\alpha > 0$ with $2|1 - 3w| < \alpha < |3 - w|$ for all w in the proposed disk. Fortunately, if $|w| < \frac{1}{7}$ then $2|1 - 3w| < 2(1 + \frac{3}{7}) = \frac{20}{7}$ and $|3 - w| > 3 - \frac{1}{7} = \frac{20}{7}$. So, $\alpha = \frac{20}{7}$ works. ■

Set 6 Problem 3: Let $f_n : \Delta \to \Delta$ be a family of biholomorphisms of Δ converging locally uniformly to a nonconstant function f. Show that f is an automorphism of Δ.

Firstly, note that because f is nonconstant (by assumption) and holomorphic (by the Weierstaß convergence), we know that $f(\Delta)$ is an open set in $\mathbb{C}$. Also, as $f_n(\Delta) \subseteq \Delta$ for all n, we can conclude that $f(\Delta) \subseteq (\overline{\Delta})^\circ = \Delta$. Thus f maps Δ to Δ. And with that we've shown that $d(f(a), f(b))$ is well-defined for all $a, b \in \Delta$.

Next note that if d is the distance function defined in the prior problem, then d is a continuous function from $\Delta \times \Delta$ to Δ. Thus for any $a, b \in \Delta$, because we have that $f(a), f(b) \in \Delta$, that $f_n(a) \to f(a)$, and that $f_n(b) \to f(b)$, we can conclude that $d(f_n(a), f_n(b)) \to d(f(a), f(b))$ as $n \to \infty$. Simultaneously though, we know by part (a) of the last problem that $d(f_n(a), f_n(b)) = d(a, b)$ for all $n \in \mathbb{N}$. Hence, this proves that $d(f(a), f(b)) = d(a, b)$ for all $a, b \in \Delta$. By part (c) of the last problem, we can thus conclude that f is a biholomorphism.

As a side note, the first paragraph is the only part of this proof where we specifically need f to be nonconstant. If f were non-constant, then you could imagine that for any $a, b \in \Delta$ we'd have that $d(f_n(a), f_n(b)) = d(a, b)$ for all n but that also $f_n(a), f_n(b)$ race towards the boundary of Δ.

For an example of this problematic behavior, consider the functions $f_n(z) = \frac{z-(1-\frac{1}{n})}{1-(1-\frac{1}{n})z}$. Each f_n is a biholomorphism of Δ. Also, we can easily see that $f_n(z)$ converges locally uniformly to -1.

After all, suppose $|z| \leq r < 1$. Then:

$$f_n(z) + 1 = \frac{z-(1-\frac{1}{n})+1-(1-\frac{1}{n})z}{1-(1-\frac{1}{n})z} = \frac{(z+1)(1-(1-\frac{1}{n}))}{1-(1-\frac{1}{n})z} = \frac{(z+1)}{n-(n-1)z}$$

It follows that:

$$|f_n(z) + 1| = \frac{|z+1|}{|n-(n-1)z|} \leq \frac{|z|+1}{n-(n-1)|z|} \leq \frac{r+1}{n-(n-1)r} = \frac{r+1}{n(1-r)+r}$$

And since $r < 1$, it follows that $\frac{r+1}{n(1-r)+r} \to 0$ as $n \to \infty$. This finishes showing that our example f_n converge locally uniformly to -1.

Set 6 Problem 4: On the upper half plane, define the pseudo-hyperbolic distance via:

$$d_{\mathfrak{h}^+}(z, w) = \left|\frac{z-w}{z-\overline{w}}\right|.$$

(a) If $C : \mathfrak{h}^+ \to \Delta$ is the Cayley transform, show that $d_\Delta(C(z), C(w)) = d_{\mathfrak{h}^+}(z, w)$ where d_Δ is the distance defined in problem 2.

$$\begin{aligned}
d_\Delta(C(z), C(w)) &= \left|\frac{\frac{z-i}{z+i} - \frac{w-i}{w+i}}{1-\left(\frac{\overline{z}+i}{\overline{z}-i}\right)\left(\frac{w-i}{w+i}\right)}\right| \\
&= \left|\frac{\frac{(w+i)(z-i)-(w-i)(z+i)}{(z+i)(w+i)}}{\frac{(\overline{z}-i)(w+i)-(\overline{z}+i)(w-i)}{(\overline{z}-i)(w+i)}}\right| \\
&= \left|\frac{\overline{z}-i}{z+i}\right| \cdot \left|\frac{w+i}{w+i}\right| \cdot \left|\frac{(w+i)(z-i)-(w-i)(z+i)}{(\overline{z}-i)(w+i)-(\overline{z}+i)(w-i)}\right| \\
&= 1 \cdot 1 \cdot \left|\frac{wz-iw+iz+1-wz-iw+iz-1}{\overline{z}w+i\overline{z}-iw+1-\overline{z}w+i\overline{z}-iw-1}\right| = \left|\frac{2i(z-w)}{2i(\overline{z}-w)}\right| = \left|\frac{z-w}{z-\overline{w}}\right| = d_{\mathfrak{h}^+}(z, w).
\end{aligned}$$

(b) Formulate the analogues of problem 2 (a) and (b) for $\mathfrak{h}^+$.

Firstly, note as a corollary to part (a) that:

$$d_{\mathfrak{h}^+}(C^{-1}(z), C^{-1}(w)) = d_\Delta((C \circ C^{-1})(z), (C \circ C^{-1})(w)) = d_\Delta(z, w).$$

Therefore, both C and C^{-1} are distance preserving. And since every holomorphic function $f : \mathfrak{h}^+ \to \mathfrak{h}^+$ can be written as $f = C^{-1} \circ g \circ C$ where $g : \Delta \to \Delta$ is a

holomorphic function (and thus contracts distance on Δ), we can thus conclude that $d_{\mathfrak{h}^+}(f(z), f(w)) \leq d_{\mathfrak{h}^+}(z, w)$ for all $z, w \in \mathfrak{h}^+$ and every holomorphic function $f : \mathfrak{h}^+ \to \mathfrak{h}^+$.

Also if f is a biholomorphism, then we know that the function $g : \Delta \to \Delta$ such that $f = C^{-1} \circ g \circ C$ is also a biholomorphism and thus preserves distance on Δ. Hence, $d_{\mathfrak{h}^+}(f(z), f(w)) = d_{\mathfrak{h}^+}(z, w)$ for all $z, w \in \mathfrak{h}^+$ and every holomorphic function $f : \mathfrak{h}^+ \to \mathfrak{h}^+$. ■

Math 200b Lecture notes:

Before I do anything else, I'll start by tracing out the isomorphism from *page 692*. Recall that because every element of $N \otimes M_i$ is a linear combination of pure tensors, it suffices to keep track of where they are sent to by each map. In a similar vein, every element of $\bigoplus_{i\in I}(N \otimes M_i)$ is a linear combination of pure tensors in each of it's coordinates. So, it suffices to keep track where those linear combinations go.

One other bit of book-keeping is that j_i is the embedding homomorphism of $N \otimes M_i$ into $\bigoplus_{i\in I}(N \otimes M_i)$. Now observe the following diagram:

$$\begin{array}{ccccc}
 & h_{\bigoplus_{i\in I}(N\otimes M_i)}(\bigoplus_{i\in I}(N \otimes M_i)) & \xrightarrow{\sim} & \prod_{i\in I} h_{N\otimes M_i}(\bigoplus_{i\in I}(N \otimes M_i)) & \\
\mathrm{id}_{\bigoplus_{i\in I}(N\otimes M_i)} \Longrightarrow & (n_i\otimes m_i)_{i\in I} \mapsto (n_i\otimes m_i)_{i\in I} & \Longmapsto & (n\otimes m_i \mapsto j_i(n\otimes m_i))_{i\in I} & \\
 & \xrightarrow{\sim} \prod_{i\in I}(h_{M_i} \circ h_N)(\bigoplus_{i\in I}(N \otimes M_i)) & & (h_{\bigoplus_{i\in I} M_i} \circ h_N)(\bigoplus_{i\in I}(N \otimes M_i) & \\
 & \Longmapsto (m_i \mapsto (n \mapsto j_i(n\otimes m_i))_{i\in I} & \Longmapsto & (m_i)_{i\in I} \mapsto (n \mapsto (n\otimes m_i)_{i\in I})) & \\
 & \xrightarrow{\sim} h_{N\otimes(\bigoplus_{i\in I} M_i)}(\bigoplus_{i\in I}(N \otimes M_i)) & & & \\
 & \Longmapsto n\otimes(m_i)_{i\in I} \mapsto (n\otimes m_i)_{i\in I} & & &
\end{array}$$

Thus to no surprise at all, in our isomorphism from $\bigoplus_{i\in I}(N \otimes M_i)$ to $N \otimes (\bigoplus_{i\in I} M_i)$ we know that $(n\otimes m_i)_{i\in I} \mapsto n\otimes(m_i)_{i\in I}$ for any fixed any $n \in N$ and any m_i. As a fairly easy corollary, if $\widehat{j}_i$ is the embedding of M_i into $\bigoplus_{i\in I} M_i$, then we'd get that $(n_i \otimes m_i)_{i\in I}$ maps to $\sum_{i\in I} n_i \otimes \widehat{j}_i(m_i)$.

Now back to the algebra we were learning in class.

Recall that for any two A-modules M_1, M_2, we have a bilinear map $\iota : M_1 \times M_2 \to M_1 \otimes M_2$ given by $(m_1, m_2) \mapsto m_1 \otimes m_2$. From that we can deduce a few arithmetic properties of the tensor product. Namely:

- $(am_1) \otimes m_2 = a(m_1 \otimes m_2) = m_1 \otimes (am_2)$ (this property is called A balance...),
- $(m_1 + m_1') \otimes m_2 = (m_1 \otimes m_2) + (m_1' \otimes m_2)$,
- $m_1 \otimes (m_2 + m_2') = (m_1 \otimes m_2) + (m_1 \otimes m_2')$.

One more note before I start covering propositions is that if there is ambiguity about what ring I'm taking a tensor product with respect to, I'll add a subscript to the $\otimes$ symbol. I specifically mention this because if M_1 and M_2 can both be interpretted as either A-modules or B-modules, then $M_1 \otimes_A M_2$ and $M_1 \otimes_B M_2$ will be different modules.

Proposition: If M is an A-module and $\mathfrak{a} \lhd A$, then $\frac{A}{\mathfrak{a}} \otimes M \cong \frac{M}{\mathfrak{a}M}$.

Proof:

Let $f : \frac{A}{\mathfrak{a}} \times M \to \frac{M}{\mathfrak{a}M}$ be defined by $f(x + \mathfrak{a}, m) = xm + \mathfrak{a}M$. To see that f is well-defined, suppose $x' = x + a$ for some $a \in \mathfrak{a}$. Then:

$$x'm = (x + a)m = xm + am \Longrightarrow x'm + \mathfrak{a}M = xm + \mathfrak{a}M.$$

Also note that f is A-bilinear.

After all, we have that

$$\begin{aligned} f(a_1(x_1 + \mathfrak{a}) + a_2(x_2 + \mathfrak{a}), m) &= f((a_1x_1 + a_2x_2) + \mathfrak{a}, m) \\ &= (a_1x_1 + a_2x_2)m + \mathfrak{a}M \\ &= a_1(x_1m + \mathfrak{a}M) + a_2(x_2m + \mathfrak{a}M) \\ &= a_1f(x_1, m) + a_2f(x_2, m) \end{aligned}$$

Meanwhile, linearity in the other argument is proven identically.

Thus by the universal property of tensor products, there must exist a unique A-module homomorphism $\widehat{f} : \frac{A}{\mathfrak{a}} \otimes M \to \frac{M}{\mathfrak{a}M}$ such that $\widehat{f}((x + \mathfrak{a}) \otimes m) = xm + \mathfrak{a}M$. To show that this is an isomorphism, we next construct an inverse homomorphism.

Define $g : M \to \frac{A}{\mathfrak{a}} \otimes M$ by $g(m) = (1 + \mathfrak{a}) \otimes m$. By the properties at the top of the page, we know that g is an A-module homomorphism. Also, we claim that $\mathfrak{a}M \subseteq \ker(g)$.

After all, note that every element of $\mathfrak{a}M$ is of the form $\sum_{i=1}^{k} a_im_i$ for some $a_i \in \mathfrak{a}$ and $m_i \in M$. So to show that $\mathfrak{a}M \subseteq \ker(g)$ it suffices to prove that $g(am) = 0$ for all $a \in \mathfrak{a}$ and $m \in M$. But this is easy.

$$g(am) = (1 + \mathfrak{a}) \otimes am = (a + \mathfrak{a}) \otimes m = (0 + \mathfrak{a}) \otimes m = 0$$

It follows that there is a well-defined A-module homomorphism $\overline{g} : \frac{M}{\mathfrak{a}M} \to \frac{A}{\mathfrak{a}} \otimes M$ given by $\overline{g}(m + \mathfrak{a}M) = (1 + \mathfrak{a}) \otimes m$.

Claim: $\widehat{f} \circ \overline{g} = \mathrm{Id}_{\frac{M}{\mathfrak{a}M}}$.

$\widehat{f} \circ \overline{g}(m + \mathfrak{a}M) = \widehat{f}((1 + \mathfrak{a}) \otimes m) = m + \mathfrak{a}M.$

Claim: $\overline{g} \circ \widehat{f} = \mathrm{Id}_{\frac{A}{\mathfrak{a}} \otimes M}$.

Since $\frac{A}{\mathfrak{a}} \otimes M$ is generated by pure tensor elements, it is enough to check the claim at a pure tensor element $(x + \mathfrak{a}) \otimes m$.

$$\overline{g}(\widehat{f}((x + \mathfrak{a}) \otimes m)) = \overline{g}(xm + \mathfrak{a}M) = (1 + \mathfrak{a}) \otimes xm = (x + \mathfrak{a}) \otimes m. \blacksquare$$

As an application of the prior theorem, we can see that if $n, m \in \mathbb{N}$ then:

$$\frac{\mathbb{Z}}{n\mathbb{Z}} \otimes_{\mathbb{Z}} \frac{\mathbb{Z}}{m\mathbb{Z}} \cong \frac{(\mathbb{Z}/m\mathbb{Z})}{n\mathbb{Z}(\mathbb{Z}/m\mathbb{Z})}$$

Yet note that $nk_1(\ell_1 + m\mathbb{Z}) + nk_2(\ell_2 + m\mathbb{Z}) = n((k_1\ell_1 + k_2\ell_2) + m\mathbb{Z})$ for any integers k_1, k_2, ℓ_1, ℓ_2. It thus follows that $n\mathbb{Z}(\mathbb{Z}/m\mathbb{Z}) \subseteq n(\mathbb{Z}/m\mathbb{Z})$ and the other inclusion is obvious.

Therefore, $\frac{(\mathbb{Z}/m\mathbb{Z})}{n\mathbb{Z}(\mathbb{Z}/m\mathbb{Z})} = \frac{\mathbb{Z}/m\mathbb{Z}}{n(\mathbb{Z}/m\mathbb{Z})} = \frac{\mathbb{Z}/m\mathbb{Z}}{(n\mathbb{Z}+m\mathbb{Z})/m\mathbb{Z}} \cong \frac{\mathbb{Z}/m\mathbb{Z}}{n\mathbb{Z}+m\mathbb{Z}} = \frac{\mathbb{Z}}{\gcd(n,m)}$.

<u>Lemma:</u> If $f_1 : M_1 \to M_1'$ and $f_2 : M_2 \to M_2'$ are A-module homomorphisms, then there is a unique A-module homomorphism $f_1 \otimes f_2 : M_1 \otimes M_2 \to M_1' \otimes M_2'$ such that $(f_1 \otimes f_2)(m_1 \otimes m_2) = f_1(m_1) \otimes f_2(m_2)$.

Proof:

Define $f : M_1 \times M_2 \to M_1' \otimes M_2'$ by $f(m_1, m_2) := f_1(m_1) \otimes f_2(m_2)$. Hopefully it's obviosu to you why f is A-bilinear. Hence, by the universal property of tensor products, there exists a unique A-module homomorphism $f_1 \otimes f_2 : M_1 \otimes M_2 \to M_1' \otimes M_2'$ doing what we want. ■

Given any A-module M, we define $T_m : A$-**mod** $\to A$-**mod** by:

$$\begin{tikzcd} N \arrow[d, "f"] \arrow[r, squiggly, "T_M"] & T_M(N) = M \otimes N \arrow[d, "T_M(f) = \mathrm{Id}_M \otimes f"] \\ N' & T_M(N') = M \otimes N' \end{tikzcd}$$

To check that T_M is a functor, first note that $\mathrm{Id}_M \otimes \mathrm{Id}_N = \mathrm{Id}_{M \otimes N}$ for all A-modules N.

We also claim that $(f_1 \otimes g_1) \circ (f_2 \otimes g_2) = (f_1 \circ f_2) \otimes (g_1 \circ g_2)$.

Since tensor products are generated by pure tensor elements, it suffices to check the above equation at pure tensor elements in order to prove it more generally.

$$\begin{aligned} ((f_1 \otimes g_1) \circ (f_2 \otimes g_2))(x \otimes y) &= (f_1 \otimes g_1)(f_2(x) \otimes g_2(y)) \\ &= f_1(f_2(x)) \otimes g_1(g_2(y)) \\ &= ((f_1 \circ f_2) \otimes (g_1 \circ g_2))(x \otimes y). \end{aligned}$$

Given an ideal $\mathfrak{a} \lhd A$, another important functor is $F_\mathfrak{a} : A$-**mod** $\to A$-**mod** defined by:

$$\begin{tikzcd} M \arrow[d, "f"] \arrow[r, squiggly, "F_\mathfrak{a}"] & F_\mathfrak{a}(M) = \frac{M}{\mathfrak{a}M} \arrow[d, "F_\mathfrak{a}(f)"] \arrow[r, equal, "(x+\mathfrak{a}M \mapsto f(x)+\mathfrak{a}M')"] & {} \\ M' & F_\mathfrak{a}(M') = \frac{M'}{\mathfrak{a}M'} & {} \end{tikzcd}$$

To see that $F_\mathfrak{a}(f)$ is well-defined, just note that $f(\mathfrak{a}M) \subseteq \mathfrak{a}M'$ as any $\sum_{i=1}^m a_i m_i \in \mathfrak{a}M$ is mapped to $\sum_{i=1}^m a_i f(m_i) \in \mathfrak{a}M'$. Hopefully it's obvious why $F_\mathfrak{a}(f \circ g) = F_\mathfrak{a}(f) \circ F_\mathfrak{a}(g)$ and why $F_\mathfrak{a}(\mathrm{Id}_M) = \mathrm{Id}_{\frac{M}{\mathfrak{a}M}}$.

Proposition: $T_{A/\mathfrak{a}} \cong F_{\mathfrak{a}}$.

Proof:

Define $\eta_M : \frac{A}{\mathfrak{a}} \otimes M \to \frac{M}{\mathfrak{a}M}$ by $(x + \overline{a}) \otimes m \mapsto xm + \mathfrak{a}M$. Then we know from two pages ago that η_M is an isomorphism for all A-modules M. Also, note for all pure tensor elements of $\frac{A}{\mathfrak{a}} \otimes M$ that:

$$\begin{tikzcd}
(x+\mathfrak{a})\otimes m \arrow[rr, mapsto] \arrow[ddd, mapsto] & & xm+\mathfrak{a}M \arrow[ddd, mapsto] \\
& \frac{A}{\mathfrak{a}} \otimes M \arrow[r, "\eta_M"] \arrow[d, "\mathrm{Id}_{\frac{A}{\mathfrak{a}}} \otimes f"] & \frac{M}{\mathfrak{a}M} \arrow[d, "\overline{f}"] \\
& \frac{A}{\mathfrak{a}} \otimes M' \arrow[r, "\eta_{M'}"'] & \frac{M'}{\mathfrak{a}M'} \\
(x+\mathfrak{a})\otimes f(m) \arrow[rr, mapsto] & & xf(m)+\mathfrak{a}M
\end{tikzcd}$$

It follows that the above diagram commutes for all elements (pure or not) of $\frac{A}{\mathfrak{a}} \otimes M$. ■

From the above proposition we can also see that T_A is naturally isomorphic to the identity functor. It follows that T_A is exact.

The following results and definitions will be incredibly similar to those on *pages 688-691*. So, I will give much less detail on proving these results.

Given an arbitrary category **C** and a collection $\{b_i\}_{i \in I}$ of objects in **C**, we say an object B equipped with a family of morphisms $\{j_i : b_i \to B\}$ is a coproduct of the b_i if and only if for any collection of morphisms $\{f_i : b_i \to a\}_{i \in I}$ there exists a unique morphism $f : B \to a$ such that below commuting diagram holds for every $i \in I$:

$$\begin{tikzcd}
B \arrow[dr, "f"] & \\
b_i \arrow[u, "j_i"] \arrow[r, "f_i"'] & a
\end{tikzcd}$$

As a side note, the prefix "co-" in a category theoretic context means that you are reversing the direction of all the arrows in your commutative diagram.

Like with products, any two coproducts of a collection $\{b_i\}_{i \in I}$ of objects in a category **C** are isomorphic. Because of this, we often denote any coproduct of $\{b_i\}_{i \in I}$ as $\coprod_{i \in I} b_i$.

Note that in the category **Set**, we can show that a disjoint union of a collection of sets is a coproduct. This is why we use the symbol $\amalg$ to denote disjoint unions.

Question: What is a coproduct of functors $\{F_i : A\text{-}\mathbf{mod} \to A\text{-}\mathbf{mod}\}_{i \in I}$?

Define $\widetilde{F}(N) = \bigoplus_{i \in I} F_i(N)$ for all $N \in \mathrm{Obj}(A\text{-}\mathbf{mod})$, and if $f : N \to N'$ is an A-module homomorphism, define $\widetilde{F}(f)$ by $(x_i)_{i \in I} \mapsto ((F_i(f))(x_i))_{i \in I}$. The proof that this is a well-defined functor is identical to the proof on *page 689*.

Also, for each i we define $j_i : F_i \to \widetilde{F}$ by setting $j_{i,N} : F_i(N) \to \widetilde{F}(N) = \bigoplus_{i' \in I} F_{i'}(N)$ to be the standard embedding homomorphism for all $N \in \mathrm{Obj}(A\text{-}\mathbf{mod})$. Then each j_i is a natural transformation. After all:

$$\begin{tikzcd}
x_i \arrow[rrr, mapsto] \arrow[dddd, mapsto] & & & j_{i,N}(x_i) \arrow[dddd, mapsto] \\
& F_i(N) \arrow[r, "j_{i,N}"] \arrow[d, "F_i(f)"] & \bigoplus_{i \in I} F_i(N) \arrow[d, "\widetilde{F}(f)"] & \\
& F_i(N') \arrow[r, "j_{i,N'}"] & \bigoplus_{i \in I} F_i(N') & \\
(F_i(f))(x_i) \arrow[rrr, mapsto] & & & j_{i,N'}((F_i(f))(x_i))
\end{tikzcd}$$

Finally, suppose $\{\eta_i : F_i \to G\}_{i \in I}$ is a a collection of natural transformations. Then for all A-modules N and $i \in I$ we have an A-module homomorphism $\eta_{i,N} : F_i(N) \to G(N)$ that is part of the natural transformation η_i.

Recall from *pages 555-556* that:

$$\prod_{i \in I} \mathrm{Hom}_A(F_i(N), G(N)) \cong \mathrm{Hom}_A(\bigoplus_{i \in I} F_i(N), G(N))$$

as A-modules via the identication of $(\phi_i)_{i \in I}$ with $(x_i)_{i \in I} \mapsto \sum_{i \in I} \phi_i(x_i)$.

Hence, for each A-module N there is a unique A-module homomorphism $\widetilde{\eta}_N : \widetilde{F}(N) \to G(N)$ satisfying that $\eta_{i,N} = \widetilde{\eta}_N \circ j_{i,N}$ for all $i \in I$. We also claim that the $\widetilde{\eta}_N$ form a natural transformation.

After all, if $f : N \to N'$ is an A-module homomorphism, consider the below diagram:

$$\begin{tikzcd}
(x_i)_{i \in I} \arrow[rrr] \arrow[ddddd, mapsto] & & & \sum_{i \in I} \eta_{i,N}(x_i) \arrow[dr, mapsto] & \\
& \bigoplus_{i \in I} F_i(N) \arrow[r, "\widetilde{\eta}_N"] \arrow[d, "\widetilde{F}(f)"'] & G(N) \arrow[d, "G(f)"] & (G(f))(\sum_{i \in I} \eta_{i,N}(x_i)) \arrow[d, equal] & \\
& & & \sum_{i \in I}((G(f)) \circ \eta_{i,N})(x_i) \arrow[dl, equal, "?"] & \\
& \bigoplus_{i \in I} F_i(N') \arrow[r, "\widetilde{\eta}_{N'}"] & G(N') & & \\
((F_i(f))(x_i))_{i \in I} \arrow[rr, mapsto] & & \sum_{i \in I}(\eta_{i,N'} \circ (F_i(f)))(x_i) & &
\end{tikzcd}$$

To prove the above diagram commutes, it suffices to note that because η_i is a natural transformation from F_i to G, we have for all $i \in I$ that:

$$\eta_{i,N'} \circ (F_i(f)) = (G(f) \circ \eta_{i,N}).$$

So, $\bigoplus_{i \in I} F_i := \widetilde{F}$ is how we define the coproduct of the functors $\{F_i\}_{i \in I}$. ■

There are two other observations I want to quickly write down.

1. If $\{a_i\}_{i \in I}$ and $\{b_i\}_{i \in I}$ are collections of objects in a category **C** such that there exists products $\prod_{i \in I} a_i$ and $\prod_{i \in I} b_i$ (equipped with the morphism $\{j_i^{(a)}\}_{i \in I}$ and $\{j_i^{(b)}\}_{i \in I}$

respectively) and if for each $i \in I$ there exists an isomorphism $\theta_i : a_i \to b_i$, then we have that $\prod_{i\in I} a_i \cong \prod_{i\in I} b_i$ via an isomorphism θ satisfying that $\theta \cdot j_i^{(a)} = j_i^{(b)} \cdot \theta_i$ for all $i \in I$.

2. If $\{F_i : A\text{-}\mathbf{mod} \to A\text{-}\mathbf{mod}\}_{i\in I}$ is a collection of functors and $G : A\text{-}\mathbf{mod} \to A\text{-}\mathbf{mod}$ is another functor, then:

$$\bigoplus_{i\in I}(F_i \circ G) = (\bigoplus_{i\in I} F_i) \circ G$$

Given an arbitrary category **C**, we can define a category $\mathbf{C}^{\mathrm{op}}$ such that $\mathrm{Obj}(\mathbf{C}^{\mathrm{op}}) = \mathrm{Obj}(\mathbf{C})$ and $\mathrm{Hom}_{\mathbf{C}^{\mathrm{op}}}(a, b) = \mathrm{Hom}_{\mathbf{C}}(b, a)$.

In particular, given any morphism $f : a \to b$ in **C**, let us temporarily denote f^{op} to be the corresponding morphism in $\mathbf{C}^{\mathrm{op}}$. If we define $f^{\mathrm{op}} \cdot g^{\mathrm{op}} = (g \cdot f)^{\mathrm{op}}$ then we have that the morphisms of $\mathbf{C}^{\mathrm{op}}$ satisfy all the necessary composition properties to be a category.

I prove this in my Aluffi notes (although I use horrible notation while doing so).

You may also note that the definition of the opposite category naturally gives rise to the following contravariant functor $\mathrm{Op} : \mathbf{C} \to \mathbf{C}^{\mathrm{op}}$:

$$\begin{tikzcd} a \arrow[d, "f"] & & a \\ b & \rightsquigarrow & b \arrow[u, "f^{\mathrm{op}}"'] \end{tikzcd}$$

Also note that $(\mathbf{C}^{\mathrm{op}})^{\mathrm{op}} = \mathbf{C}$.

Lemma: Any contravariant functor $F : \mathbf{C} \to \mathbf{D}$ can be written as $G \circ \mathrm{Op}$ where $G : \mathbf{C}^{\mathrm{op}} \to \mathbf{D}$ is a functor.

Simply define $G(f^{\mathrm{op}}) = F(f)$ for all $f \in \mathrm{Hom}_{\mathbf{C}}(a, b)$ and objects $a, b \in \mathrm{Obj}(\mathbf{C})$.

Similarly, given any functor $G : \mathbf{C}^{\mathrm{op}} \to \mathbf{D}$ we have that $G \circ \mathrm{Op}$ is a contravariant functor from **C** to **D**.

Hopefully this is obvious.

A natural transformation $\eta : F_1 \to F_2$ between two contravariant functors from **C** to **D** would be a collection of homomorphisms $\eta_N : F_1(N) \to F_2(N)$ such that the below diagram commutes for all morphisms $f : a \to b$ in **C**:

$$\begin{tikzcd} F_1(a) \arrow[r, "\eta_a"] & F_2(a) \\ F_1(b) \arrow[u, "F_1(f)"] \arrow[r, "\eta_b"'] & F_2(b) \arrow[u, "F_1(f)"'] \end{tikzcd}$$

Clearly if G_1, G_2 are functors from $\mathbf{C}^{\mathrm{op}}$ to **D** such that $F_i = G_i \circ \mathrm{Op}$, then η is a natural transformation (or natural isomorphism) from $F_1 \to F_2$ iff η is a natural transformation from G_1 to G_2.

Consequently, we have an invertible functor from the category of contravariant functors from **C** to **D** (which I realize I've never properly defined) to the category of functors from $\mathbf{C}^{\mathrm{op}}$ to **D**.

Hopefully this gives some justification on why we can define products and coproducts of contravariant functors from A-**mod** to A-**mod** the way we do.

Our product is:

- $(\prod_{i\in I} F_i)(N) = \prod_{i\in I} F_i(N)$
- $(\prod_{i\in I} F_i)(f : N \to N')$ equals the map $\prod_{i\in I} F_i(N') \to \prod_{i\in I} F_i(N)$ given by $(x_i)_{i\in I} \mapsto ((F_i(f))(x_i))_{i\in I}$.

Our coproduct is similarly defined as $\bigoplus_{i\in I} F_i$

Also, hopefully this helps explain why all our prior results about products and coproducts of functors also apply to contravariant functors.

Observation: Suppose $\{M_i\}_{i\in I}$ is a collection of A-modules. Then $\prod_{i\in I} h^{M_i} \cong h^{\prod_{i\in I} M_i}$.

Proof:

Recall from *pages 555-556* that:

$$\prod_{i\in I} h^{M_i}(N) = \prod_{i\in I} \mathrm{Hom}_A(N, M_i) \cong \mathrm{Hom}_A(N, \prod_{i\in I} M_i) = h^{\prod_{i\in I} M_i}(N)$$

via the isomorphism sending $(\phi_i)_{i\in I}$ to the map $x_i \mapsto (\phi_i(x_i))_{i\in I}$. Similarly to on *page 691*, you can also see that that this defines a natural transformation. ■

Due to issues along the lines of Russell's paradox, we can not have a category of all categories. That said, we can create a category **Cat** of small categories where morphisms are functors between those categories. Then, hopefully any definitions or insights in this restricted category can extend beyond it.

As a side note, **C** is called a small category iff $\mathrm{Obj}(\mathbf{C})$ is a set. Many of the categories we care about (such as **Set**) are not small categories. That said, for an example of a small category see *page 618*.

Question: What is a product of two categories $\mathbf{C}_1$ and $\mathbf{C}_2$ in **Cat**?

Let $\mathrm{Obj}(\mathbf{D}) = \{(a_1, a_2) : a_1 \in \mathrm{Obj}(\mathbf{C}_1) \text{ and } a_2 \in \mathrm{Obj}(\mathbf{C}_2)\}$ and for all $(a_1, a_2), (b_1, b_2)$ in $\mathrm{Obj}(\mathbf{D})$ define $\mathrm{Hom}_{\mathbf{D}}((a_1, a_2), (b_1, b_2))$ to be the class of all (f, g) where $f \in \mathrm{Hom}_{\mathbf{C}_1}(a_1, b_1)$ and $g \in \mathrm{Hom}_{\mathbf{C}_2}(a_2, b_2)$. Also define morphism composition to be pairwise. Then **D** is easily checked to be a small category.

Next, for each $i \in I$ define the functor $\pi_i : \mathbf{D} \to \mathbf{C}_i$ such that $\pi_i((a_1, a_2)) = a_i$ and $\pi_i((f_1, f_2)) \mapsto f_i$. It is easily checked that each π_i is a well-defined functor.

Finally, suppose $F_1 : \mathbf{E} \to \mathbf{C}_1$ and $F_2 : \mathbf{E} \to \mathbf{C}_2$ are functors. Then we want to show that there is a unique functor $\widetilde{F} : \mathbf{E} \to \mathbf{D}$ such that $F_i = \pi_i \circ \widetilde{F}$ for both i. Yet note that $\widetilde{F}(a)$ must equal $(F_1(a), F_2(a))$ for all $a \in \mathrm{Obj}(\mathbf{E})$ in order for $\pi_i \circ \widetilde{F} = F_i$ for both i. Similarly, we must have that $(f : a \to b) \mapsto (F_1(f), F_2(f))$ for all morphisms $f : a \to b$ in **E**.

Upon checking that $\widetilde{F}$ defined as we said was required is in fact a functor, this proves that **D** is a product of $\mathbf{C}_1$ and $\mathbf{C}_2$. ■

Importantly, no part of the prior construction fails even if $\mathbf{C}_1$ and $\mathbf{C}_2$ aren't small categories. So, in general, we define $\mathbf{C}_1 \times \mathbf{C}_2$ such that:

- its class of objects the class of all pairs of objects (a_1, a_2) where $a_1 \in \mathrm{Obj}(\mathbf{C}_1)$ and $a_2 \in \mathrm{Obj}(\mathbf{C}_2)$;
- $\mathrm{Hom}_{\mathbf{C}_1 \times \mathbf{C}_2}((a_1, a_2), (b_1, b_2))$ is the collection of all pairs of morphisms (f, g) where $f : a_1 \to b_1$ is a morphism in $\mathrm{Hom}_{\mathbf{C}_1}(a_1, b_1)$ and $g : a_2 \to b_2$ is a morphism in $\mathrm{Hom}_{\mathbf{C}_2}(a_2, b_2)$;
- morphism composition is pairwise.

Set 7 Problem 3: For given functors $F_1, f_2 : \mathbf{C} \to \mathbf{C}$, defin

Lemma: If G is a simply connected region and $f : G \to \mathbb{C}$ is a holomorphic function, then for any $n \in \mathbb{N}$ there exists $g \in O(G)$ such that $g^n = f$.

By a theorem on *page 416*, we know there exists a holomorphic function $u : G \to \mathbb{C}$ such that $f(z) = e^{u(z)}$. Now just define $f(z) = e^{u(z)/n}$.

If a region G satisfies that any nonzero holomorphic function admits a holomorphic square root function, we say G is a root domain. The above lemma tells us that any simply connected region is a root domain.

(We'll later show that every root domain is simply connected as well. See *page* ____.)

The Riemann Mapping Theorem: If $G \neq \mathbb{C}$ is a root domain, then U is biholomorphic to Δ.

Proof:
Pick any $a \in G$.

Our first goal will be to show that there exists an injective holomorphic function $f : G \to \Delta$ such that $f(a) = 0$. That way, we can say that the family of functions $\mathcal{F} := \{f : G \to \Delta : f \text{ is injective, holomorphic, and } f(a) = 0\}$ is nonempty.

As a side note, in the original German literature these functions are called Schlicht functions.

Pick $b \in G^{\mathsf{C}}$ and define $h(z) = z - b$ as a function from G to $\mathbb{C}$. Note that h is a nowhere-zero holomorphic function in G. Hence, there exists a function $g : G \to \mathbb{C}$ with $h = g^2$.

Claim 1: g is injective.

Suppose $g(z_1) = g(z_2)$. Then $z_1 - b = g(z_1)^2 = g(z_2)^2 = z_2 - b \implies z_1 = z_2$.

Claim 2: $g(G) \cap (-g)(G) = \emptyset$.

Suppose for the sake of contradiction that there exists $z_1, z_2 \in G$ with $g(z_1) = -g(z_2)$. Then $g(z_1)^2 = g(z_2)^2 \implies z_1 = z_2$. So, we've shown that $g(z_1) = -g(z_1)$. That in turn implies that $g(z_1)^2 = 0$. So, $h(z_1) = g(z_1)^2 = 0$. But now we have a contradiction as $h \neq 0$ everywhere on G.

Claim 3: There exists $c \in \mathbb{C}$ and $r > 0$ such that $|g(z) - c| > r$ on G.

By the open mapping theorem, we know that $(-g)(G)$ is an open set in $\mathbb{C}$. So, $\exists \overline{\Delta}(c, r) \subseteq (-g)(G)$. In turn, by claim 2 we have that $g(G) \cap \overline{\Delta}(c, r) = \emptyset$.

Let $\widetilde{f}(z) = \frac{r}{g(z) - c}$. By claim 3 we know that $\widetilde{f}$ mapso G into Δ. Also because g is injective, we know that $\widetilde{f}$ is as well.

Finally, set $f(z) = \frac{1}{2}(\widetilde{f}(z) - \widetilde{f}(a))$. By the convexity of Δ, we know that f maps G into Δ. Also f is injective because $\widetilde{f}$ is. And finally, $f(a) = 0$.

Our next goal will be to find a function $f \in \mathcal{F}$ with a maximal value for $|f'(a)|$. For some intuition on why'd we want to do this, note that if $f \in \mathcal{F}$ is to be surjective, then we'll need f to expand out each "infinitesimally small" patch of area as much as possible, and from a basic calculus class we know that expansion is proportional to the derivative of f.

Set $\mathcal{F} := \{f : G \to \Delta : f$ is injective, holomorphic, and $f(a) = 0\}$. Then set $M = \sup_{f \in \mathcal{F}} |f'(a)|$. (As we will show, M is finite. That said, for now we will allow for M to be infinite.)

We can find a sequence $\{f_n\}_{n \in \mathbb{N}}$ with $|f_n'(a)| \to M$ as $n \to \infty$. But note that $\mathcal{F}$ is bounded by 1 on G. Therefore, by Montel's theorem we have that $\mathcal{F}$ is normal. In turn, by passing to a subsequence we may assume that $f_n \xrightarrow{\ell.u.} f$ for some $f \in O(G)$.

Note that $f_n(a) = 0$ for all n implies that $f(a) = 0$. Also by the Weierstraß convergence theorem, we know that $f_n'(a) \to f'(a)$ as $n \to \infty$. This proves that $|f'(a)| = M$. (which also shows M is finite). Thirdly, by our second corollary to Hurwitz's theorem (see *page 530*) we know that either f is injective or f is constant. But by the following reasoning, we can conclude f is not constant. So, f must be injective.

If f is constant, then we must have that $f'(a) = 0$. Yet that would imply $M = 0$ and therefore $g'(a) = 0$ for all $g \in \mathcal{F}$. This contradicts that each $g \in \mathcal{F}$ is injective.

Fourthly, since f is injective (and thus nonconstant), we know by the open mapping theorem that $f(G)$ is open. Also, as $f_n(z) \in \Delta$ for all $n \in \mathbb{N}$ and $z \in G$, we can conclude that $f(G) \subseteq \overline{\Delta}$. Putting the prior two observations together, we have that $f(G) \subseteq (\overline{\Delta})^\circ = \Delta$. So, f maps G into Δ. This proves that $f \in \mathcal{F}$.

Our third and final goal will be to show that if $f \in \mathcal{F}$ satisfies that $|f'(a)|$ is maximal, then f is a surjection.

Set 6 Problem 5: Let $G \subseteq \mathbb{C}$ be a simply connected region with $G \neq \mathbb{C}$.

(a) Show that any holomorphic map $f : G \to G$ other than the identity admits at most one fixed point.

Suppose for the sake of contradiction that f has two distinct fixed points $a, b \in G$. By the Riemann mapping theorem, we know there is a biholomorphism $\theta : G \to \Delta$. In turn, we know that $g := \theta \circ f \circ \theta^{-1}$ is a holomorphic function from Δ to Δ. Also, $g := \theta \circ f \circ \theta^{-1}$ has two distinct fixed points $\alpha := \theta(a)$ and $\beta := \theta(b)$. If we can prove that g is the identity map, then we will also know that f is the identity map.

Given any $z_0 \in \Delta$, let $\varphi_{z_0}(z)$ denote the biholomorphism of the form $\frac{z-z_0}{1-\overline{z_0}z}$. Then note that $h := \varphi_\alpha \circ g \circ \varphi_{-\alpha}$ is a holomorphism of Δ with the distinct fixed points 0 and $\varphi_\alpha(\beta)$. By Schwarz's lemma, we can thus conclude that h is a rotation. Yet the only rotation with a fixed point besides 0 is the identity map. So, h is the identity map. In turn, so is g.

(b) Show that the above statement is false for $G = \mathbb{C}$.

Let $f(z) = z^2$. Then f is a holomorphic map from $\mathbb{C}$ to $\mathbb{C}$ with 0 and 1 as fixed points. This contradicts part (a).

(c) Now assume $f : G \to G$ is holomorphic and admits a fixed point a such that $f'(a) = 1$. Show that f must be the identity.

Let $\theta : G \to \Delta$ be a biholomorphic function sending a to 0. Then note that $g = \theta \circ f \circ \theta^{-1}$ is a holomorphic function from Δ to Δ such that $g(0) = 0$. Also note that:

$$\begin{aligned} g'(0) &= \theta'(f(\theta^{-1}(0))) \cdot f'(\theta^{-1}(0)) \cdot (\theta^{-1})'(0) \\ &= \theta'(a) \cdot f'(a) \cdot (\theta^{-1})'(0) = \theta'(a) \cdot f'(a) \cdot \frac{1}{\theta'(\theta^{-1}(0))} = \frac{\theta'(a)}{\theta'(a)} \cdot f'(a) = 1 \end{aligned}$$

It follows by Schwarz' lemma that g is a rotation. Also, since the only rotation with $g'(0) = 1$ is the identity map, we know that g is the identity map. This in turn proves that f is also the identity map. ■